concordance

August 26, 2025

1 Concordance and Performance of IVFPQ vs. KNN

SCimilarity uses TileDB KNN IVF_FLAT index which provides exact vector similarity search.

Cytoverse quantizes the vectors via Product Quantization (PQ) and then performs an approximate lookup based on partitions in an Inverted File Index (IVF) fetched over HTTP.

Note: The memory calculations use tracemalloc.test_traced_memory() and are only valid from a fresh run of this notebook on a restarted kernel. We call gc.collect() to attempt a clean measurement between each set of parameters

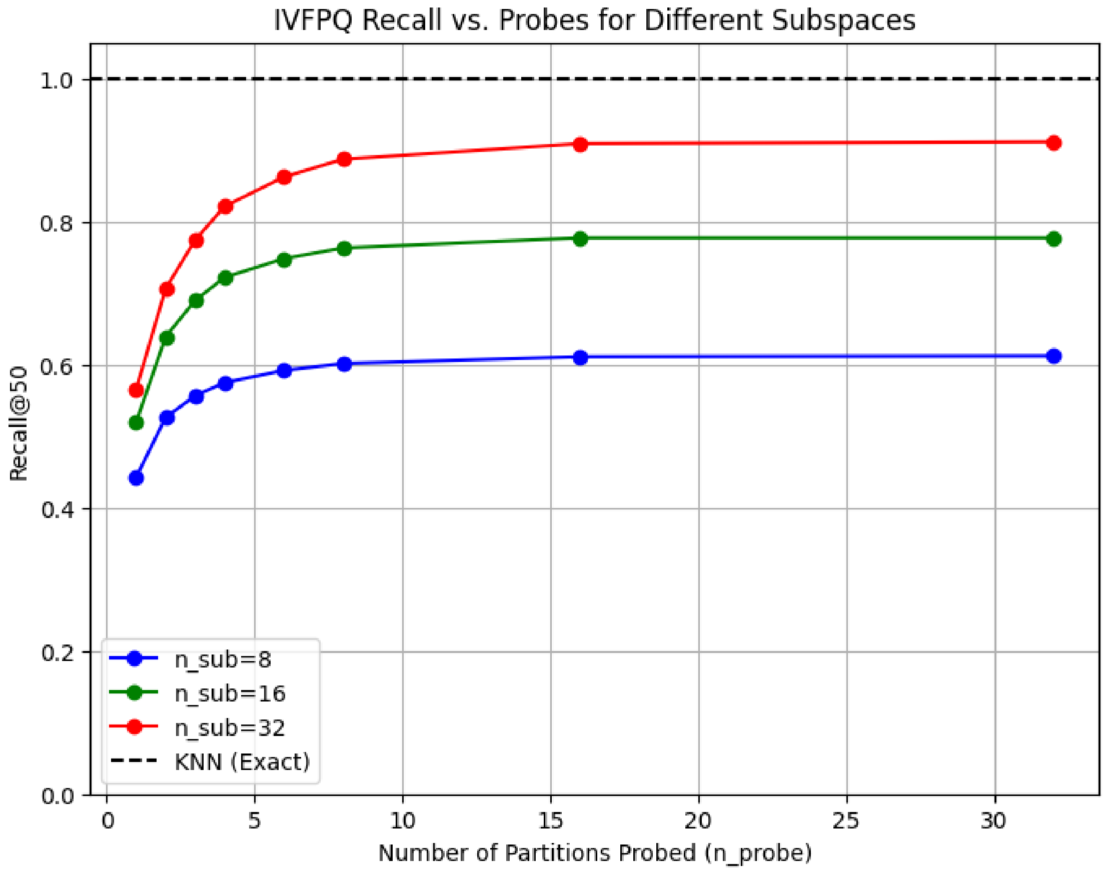
```
KNN execution time: 42.8127 seconds
KNN memory (~bandwidth) usage: 1042.91 MiB
KNN peak memory usage: 1042.92 MiB
KNN on-disk storage: 9023.19 MiB
```

1.1 Summary

Method	n_sub	n_probe	Recall	Time (s)	Speedup	Storage	Reduction	BW
IVFPQ	8	1	0.442	1.8	24.0×	92 MB	98×	1.2 MB
IVFPQ	8	2	0.526	4.3	9.9×	92 MB	98×	2.3 MB
IVFPQ	8	3	0.556	6.2	6.9×	92 MB	98×	3.3 MB
IVFPQ	8	4	0.575	8.5	5.1×	92 MB	98×	4.4 MB
IVFPQ	8	6	0.592	13.0	3.3×	92 MB	98×	6.7 MB
IVFPQ	8	8	0.601	18.4	2.3×	92 MB	98×	8.4 MB
IVFPQ	8	16	0.611	39.3	1.1×	92 MB	98×	15.2 MB
IVFPQ	8	32	0.612	74.7	0.6×	92 MB	98×	26.0 MB
IVFPQ	16	1	0.519	1.9	23.2×	153 MB	59×	2.0 MB
IVFPQ	16	2	0.639	4.2	10.2×	153 MB	59×	3.7 MB
IVFPQ	16	3	0.689	6.3	6.8×	153 MB	59×	5.5 MB
IVFPQ	16	4	0.722	8.5	5.0×	153 MB	59×	7.4 MB
IVFPQ	16	6	0.748	13.8	3.1×	153 MB	59×	11.1 MB
IVFPQ	16	8	0.763	17.7	2.4×	153 MB	59×	14.1 MB
IVFPQ	16	16	0.777	39.9	1.1×	153 MB	59×	25.2 MB
IVFPQ	16	32	0.777	75.2	0.6×	153 MB	59×	43.4 MB
IVFPQ	32	1	0.565	1.9	23.2×	274 MB	33×	3.7 MB
IVFPQ	32	2	0.706	4.2	10.2×	274 MB	33×	6.8 MB
IVFPQ	32	3	0.774	6.6	6.5×	274 MB	33×	10.0 MB
IVFPQ	32	4	0.821	8.9	4.8×	274 MB	33×	13.3 MB
IVFPQ	32	6	0.862	13.2	3.3×	274 MB	33×	20.1 MB
IVFPQ	32	8	0.887	18.6	2.3×	274 MB	33×	25.5 MB
IVFPQ	32	16	0.908	41.6	1.0×	274 MB	33×	45.5 MB
IVFPQ	32	32	0.911	79.7	0.5×	274 MB	33×	77.9 MB
KNN	**-**	**-**	**1.000**	**42.9**	**1.0×**	**9023 MB**	**1.0×**	**0.0 MB**

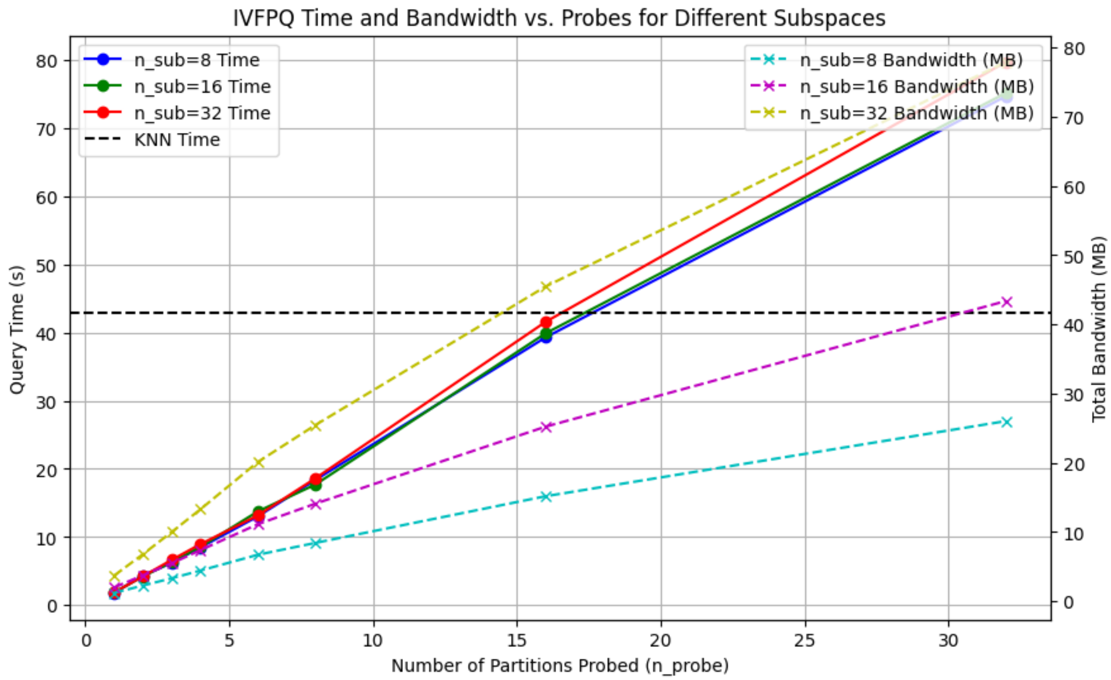
1.2 Recall vs. Probes

IVFPQ recall improves with more probes but plateaus. Fewer subspaces trade accuracy for compression reducing server storage and bandwidth.



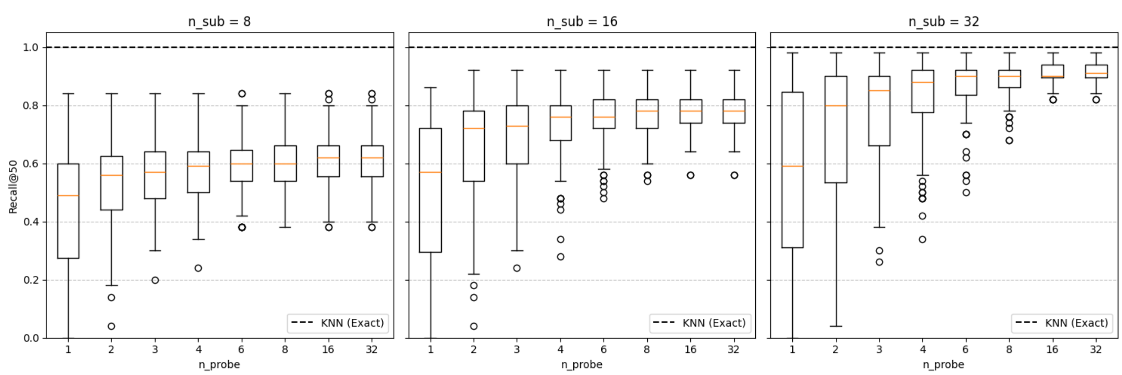
1.3 Time and Bandwidth vs. Probes

Shows tradeoff between PQ quantization and its impact on overall bandwidth and time vs. number of partitions probed. IVFPQ queries are faster than KNN, with bandwidth scaling linearly with probes



1.4 Recall@K by IVGPQ Parameter

Distribution of Recall@K across all queries for each n_probe value grouped by n_sub. Shows variability in recall (not just the mean) across queries, helping researchers assess consistency for single-cell RNA-seq labeling in a web app vs. local KNN.



1.5 Distance Distortion vs. Recall@K

Scatter plots of the relationship between mean distance distortion (for matched neighbors) and Recall@K for each query, with points colored by the average nearest neighbor distance in single-cell RNA-seq data (nn_dist_SC, reflecting query "difficulty"). This helps researchers understand if queries with farther neighbors (harder cases) exhibit higher distortion or lower recall in IVFPQ compared to KNN, relevant for web app vs. local tool tradeoffs.